

Relpart Decomposition Algorithms in UHF-SQL

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Let R be a relation composed of one key attribute denoted k and one non-key attribute denoted a . More generally, let n be the total number of non-key attributes in a relation.

1. UHF

The algorithm generates a historicized schema in 6NF, where the key induces the creation of a key partition and each attribute induces an attribute partition.

1.1. Valid-time unitemporal schema

Table 1. Naming of valid-time unitemporal partitions.

Category	Alias type:domain	Relpart identifier
Vbx	since:Vbx_POINT	<id_relvar>_since
Vbe	during:Vbe_INTERVAL	<id_relvar>_during
Vxe	until:Vxe_POINT	<id_relvar>_until

General formula: R is partitioned into $3 + 3 \times n$ relparts (where n represents the number of non-key attributes):

- **3 partitions:** $R@Vbx$, $R@Vbe$ and $R@Vxe$, each containing a corresponding valid-time attribute.

- n partitions $R_{a_i}@Vbx$, n partitions $R_{a_i}@Vbe$ and n partitions $R_{a_i}@Vxe$. Each partition contains the key attributes of R , the attribute a_i , and a valid-time attribute corresponding to a_i .

The following figure illustrates an example of the valid-time unitemporalization of a relation according to the UHF model.

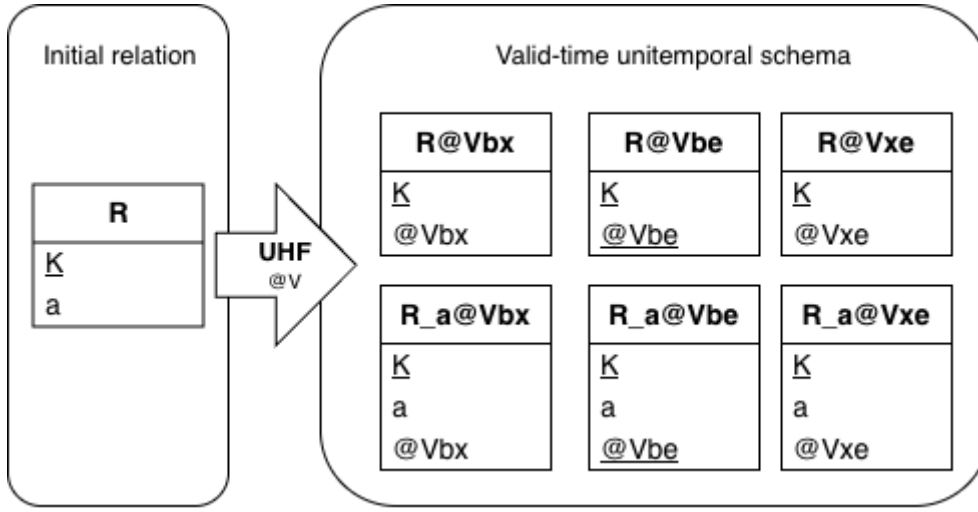


Figure 1. Relparts according to the valid-time category.

1.2. Transaction-time unitemporal schema

Table 2. Naming of transaction-time unitemporal partitions.

Category	Alias type:domain	Relpart identifier
Tbx	created:Tbx_POINT	<id_relvar>_created
Tbe	lifespan:Tbe_INTERVAL	<id_relvar>_lifespan

General formula: R is partitioned into $2 + 2 \times n$ relparts (where n represents the number of non-key attributes):

- **2 partitions:** $R@Tbe$ and $R@Tbx$, each containing a corresponding valid-time attribute.
- 2 partitions $R_{a_i}@Tbe$ and $R_{a_i}@Tbx$ for each non-key attribute a_i , each containing the key attributes of R , the attribute a_i , and a transaction-time temporal attribute for a_i .

The following figure illustrates an example of the transaction-time unitemporalization of a relation according to the UHF model.

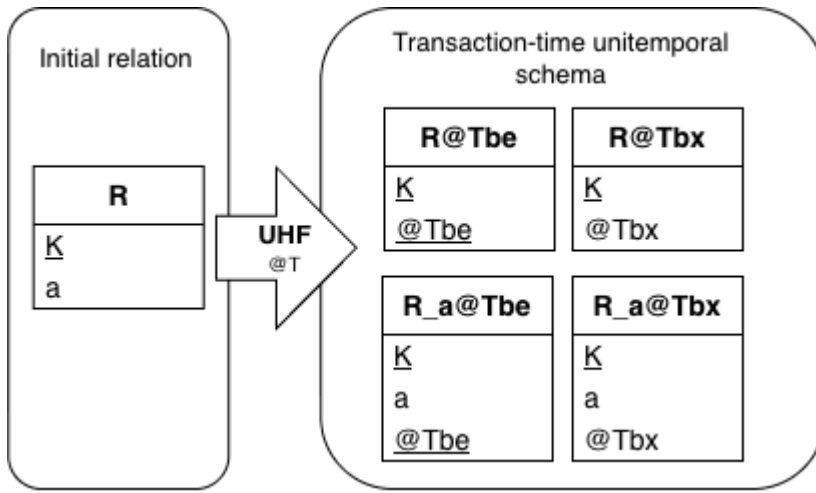


Figure 2. Relparts according to the transaction-time category.

1.3. Bitemporal schema

General formula: **R** is partitioned into $6 + 6 \times n$ relparts (where n represents the number of non-key attributes):

- **6 partitions:** $R@VbxTbx$, $R@VbxTbe$, $R@VbeTbx$, $R@VbeTbe$, $R@VxeTbx$ and $R@VxeTbe$, each containing the key attributes of **R**, a valid-time temporal attribute, and a transaction-time temporal attribute.
- **n partitions:** $R_{a_i}@VbxTbx$, n partitions $R_{a_i}@VbxTbe$, n partitions $R_{a_i}@VbeTbx$, n partitions $R_{a_i}@VbeTbe$, n partitions $R_{a_i}@VxeTbx$ and n partitions $R_{a_i}@VxeTbe$, each containing the key attributes of **R**, the attribute a_i , a valid-time temporal attribute, and a transaction-time temporal attribute.

The following figure illustrates an example of the bitemporalization of a relation according to the UHF model.

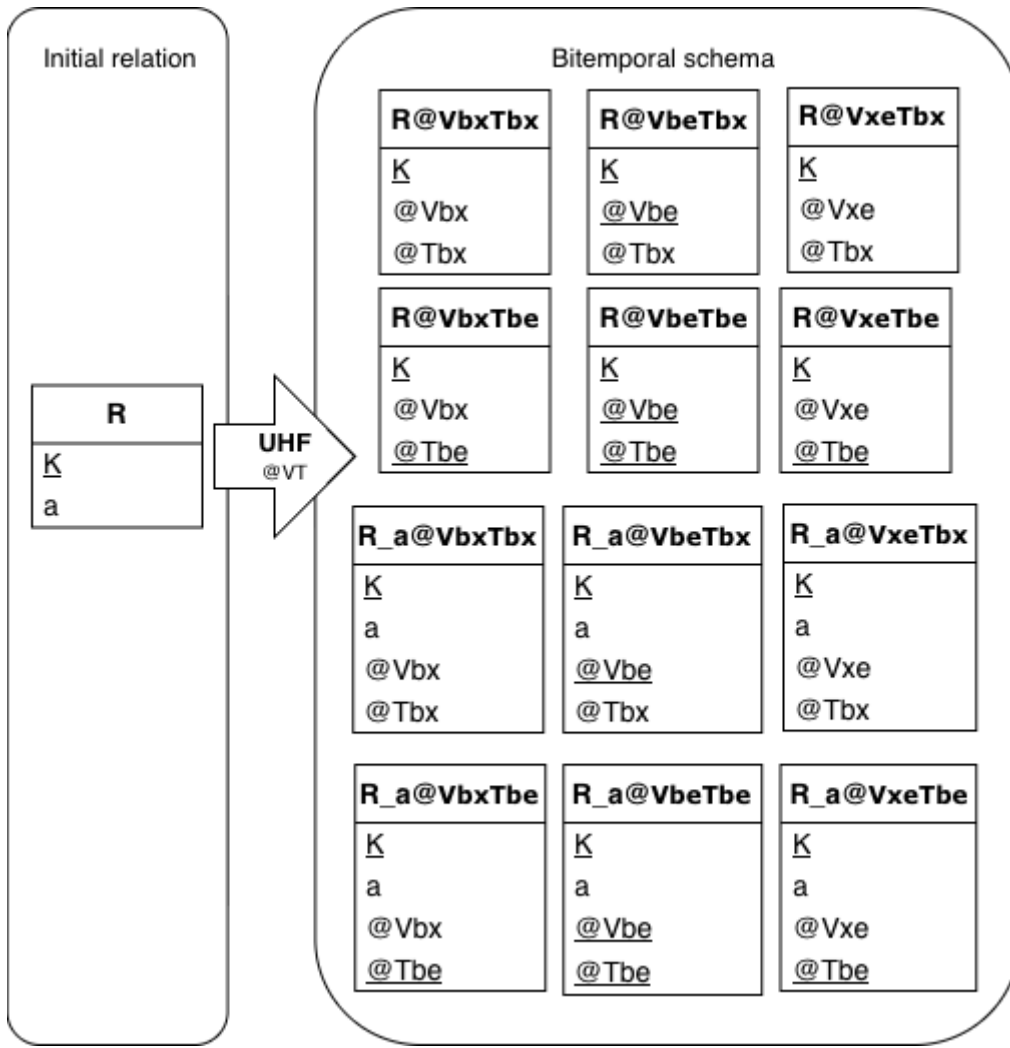


Figure 3. Relparts according to the bitemporal category.

2. DDLM

2.1. Valid-time unitemporal schema

The algorithm generates a historicized schema where the key and each attribute induce a partition in 6NF for the @Vbe category. For the @Vbx category, for each relvar (source relation) the algorithm generates a single partition in 5NF containing the key attributes, a temporal attribute since for the key, and each non-key attribute a_i with its corresponding temporal attribute a_{i_since} .

General formula: R is partitioned into $2 + n$ relparts (where n represents the number of non-key attributes):

- One partition $R@Vbx$ containing all attributes of R , a valid-time temporal attribute for each non-key attribute denoted a_{i_since} , and a valid-time temporal attribute for the key attributes denoted $@Vbx$.
- One partition $R@Vbe$ containing the key attributes of R and a valid-time temporal attribute $@Vbe$.
- n partitions $R_{a_i}@Vbe$ for each non-key attribute a_i .

The following figure illustrates an example of the valid-time unitemporalization of a relation

according to the DDLM model.

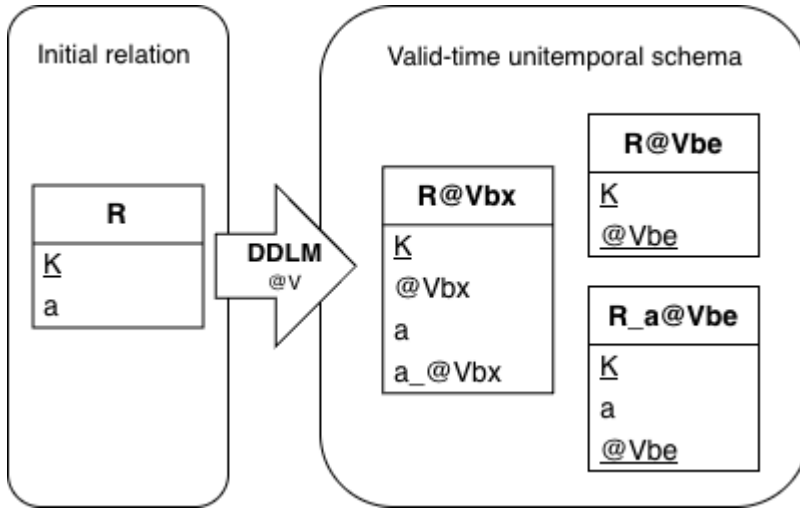


Figure 4. Relparts according to the valid-time category.

2.2. Transaction-time unitemporal schema

The algorithm generates a historicized schema in 6NF, where the key induces the creation of a key partition and each attribute induces an attribute partition.

General formula: R is partitioned into n relparts (where n represents the number of non-key attributes):

- One partition $R@Tbe$ containing the key attributes of R and a transaction-time temporal attribute $@Tbe$.
- n partitions $R_{a_i}@Tbe$ for each non-key attribute a_i .

The following figure illustrates an example of the transaction-time unitemporalization of a relation according to the DDLM model.

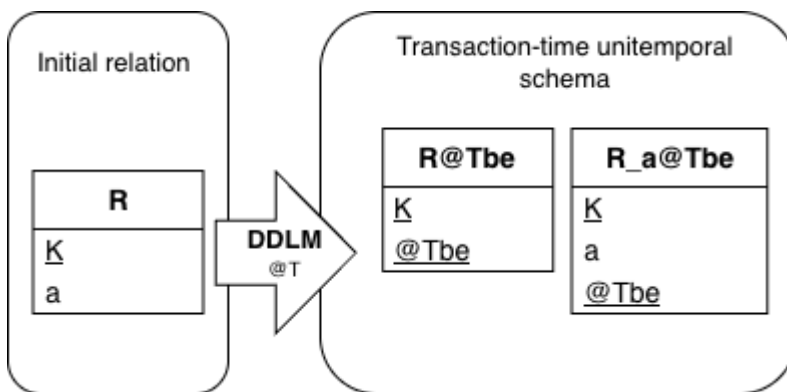


Figure 5. Relparts according to the transaction-time category.

2.3. Bitemporal schema

The algorithm generates a historicized schema where the key and each attribute induce a partition in 6NF containing both $@Vbe$ and $@Tbe$ temporal attributes. For the $@Vbx$ category, for each relvar (source relation) the algorithm generates a single partition in 5NF containing the key attributes, a

temporal attribute since for the key, each non-key attribute a_i with its corresponding temporal attribute a_{i_since} , and a @Tbe temporal attribute.

General formula: R is partitioned into $2 + n$ relparts (where n represents the number of non-key attributes):

- One partition $R@VbxTbe$ containing all attributes of R , a valid-time temporal attribute $@Vbx$ for each non-key attribute ($R_{a_i_@Vbx}$), a valid-time temporal attribute $@Vbx$ for the key attributes, and a transaction-time temporal attribute $@Tbe$.
- One partition $R@VbeTbe$ containing the key attributes of R , a valid-time temporal attribute $@Vbe$, and a transaction-time temporal attribute $@Tbe$.
- n partitions $R_{a_i_@VbeTbe}$, each containing the key attributes of R , the attribute a_i , a valid-time temporal attribute $@Vbe$, and a transaction-time temporal attribute $@Tbe$.

The following figure illustrates an example of the bitemporalization of a relation according to the DDLM model.

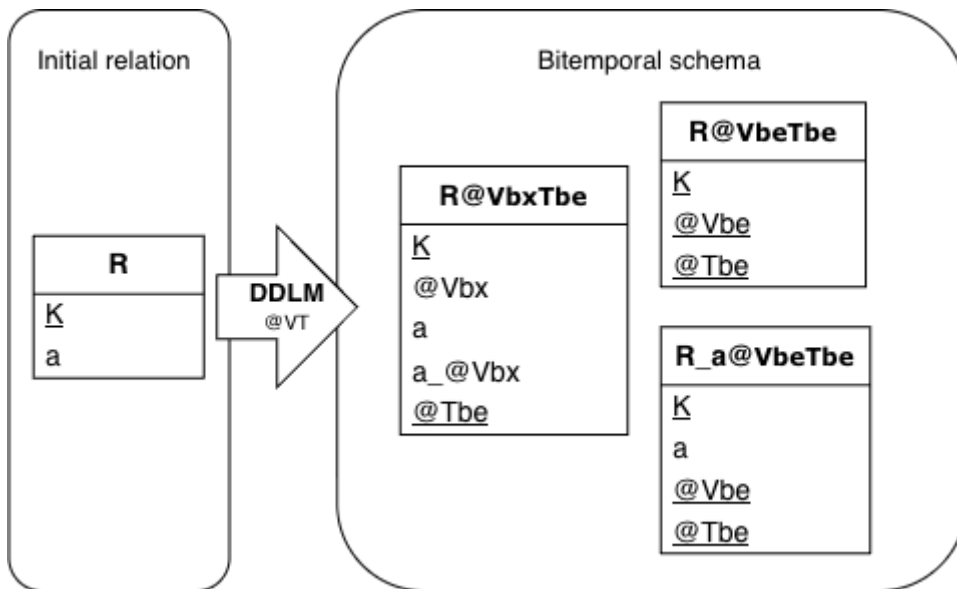


Figure 6. Relparts according to the bitemporal category.

3. BCDM

The algorithm generates a historicized schema by adding a single temporal attribute ($@Vbe$, $@Tbe$, $@Tbx$, $@VbeTbx$ or $@VbeTbe$) to each source relation depending on the target temporal category.

3.1. Valid-time unitemporal schema

General formula: R is partitioned into **one** relpart $R@Vbe$ containing all attributes of R , with the addition of a valid-time temporal attribute denoted $@Vbe$.

The following figure illustrates an example of the valid-time unitemporalization of a relation according to the BCDM model.

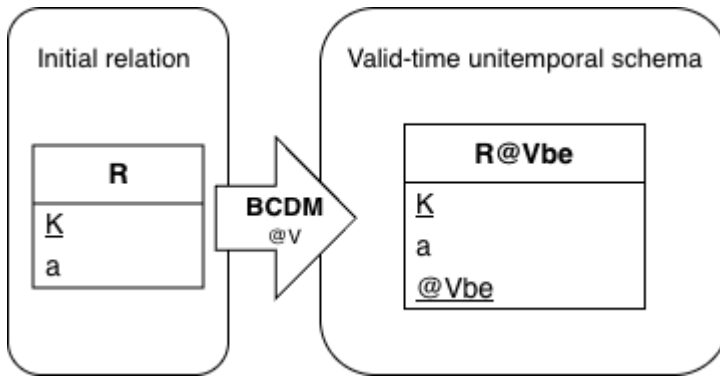


Figure 7. Relparts according to the valid-time category.

3.2. Transaction-time unitemporal schema

General formula: R is partitioned into 2 relparts:

- One partition $R@Tbe$ containing all attributes of R and a transaction-time temporal attribute $@Tbe$.
- One partition $R@Tbx$ containing all attributes of R and a transaction-time temporal attribute $@Tbx$.

The following figure illustrates an example of the transaction-time unitemporalization of a relation according to the BCDM model.

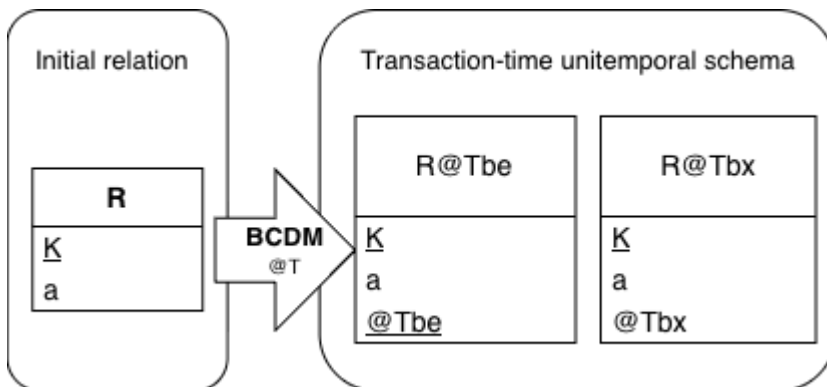


Figure 8. Relparts according to the transaction-time category.

3.3. Bitemporal schema

General formula: R is partitioned into 2 relparts:

- One partition $R@VbeTbe$ containing all attributes of R , a valid-time temporal attribute $@Vbe$, and a transaction-time temporal attribute $@Tbe$.
- One partition $R@VbeTbx$ containing all attributes of R , a valid-time temporal attribute $@Vbe$, and a transaction-time temporal attribute $@Tbx$.

The following figure illustrates an example of the bitemporalization of a relation according to the BCDM model.

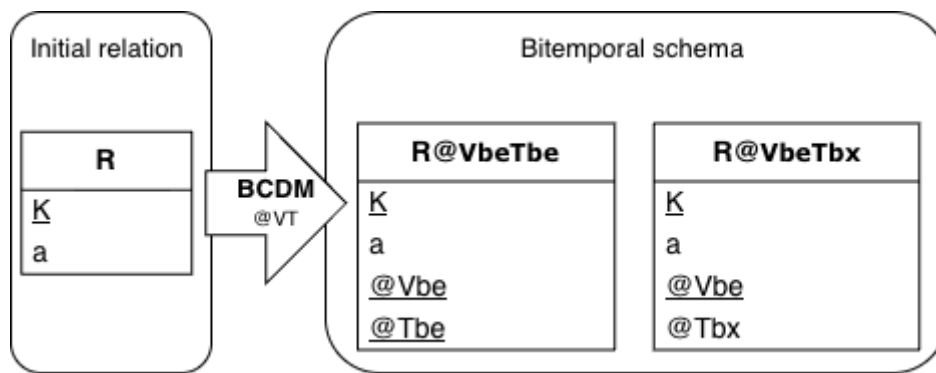


Figure 9. Relparts according to the bitemporal category.